Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Mixed Signals
Project / Solution Name	Glacier Link
Name (Leader)	Soh Ning Xuan
Email & Phone (Leader)	ninxuannnn06@gmail.com, 012-6370306
University Affiliation(s)	University of Nottingham Malaysia, University of Southampton Malaysia, Monash University Malaysia
Live Prototype URL	https://sdc-2026-mixed-signals-preliminary.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/xuan-066/SDC2026_Mixed-Signals_Preliminary.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

[Maximum 200 words]

Glacier Link's staff and managers use a React, TypeScript and Vite interface that connects to a FastAPI REST API. The API uses SQLModel to manage data stored in SQLite by default. Cold-chain rules evaluate temperature readings, permissions, approvals and recovery status. When LLM_ENABLED=true, an optional Claude parser can interpret free-text reports; if it encounters an API error, refusal or invalid output, deterministic rules provide a fallback. Clerk handles authentication in normal mode, while demo mode simulates users and roles. Notifications and incident updates appear in the webpage UI; the application does not send WhatsApp messages. The repository documents a Replit deployment path, but a live deployment has not been verified. All sample businesses and records are synthetic.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12; TypeScript/JavaScript; HTML; SQL	Backend, UI, persistence	Project code; verify component license files
FastAPI 0.156.4; SQLModel 0.0.48; Uvicorn 0.54.0	API, data models, server	Verify pinned dependency licenses
React; Vite; Tailwind CSS	Web interface/build/styling	Verify package metadata and license notices
@clerk/react 6.18.0; @clerk/themes 2.4.57	Authentication UI/provider	Clerk service terms + package licenses
Anthropic Python SDK 1.17.0; httpx 0.28.1	Optional Claude API integration / HTTP	Verify exact package license and provider terms
pytest 9.1.1; PyJWT[crypto] 2.15.1	Tests / JWT verification	Verify pinned dependency licenses

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Anthropic Messages API	Optional free-text report parser; only if LLM_ENABLED=true	Paid/usage-based or applicable account tier; team to confirm
Clerk API / hosted auth	Authentication in non-demo mode	Plan/region to confirm
No external messaging API	Notifications are shown in the webpage UI; no WhatsApp or other external messaging service is used	Not applicable

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Deployment instructions documented; actual live deployment not verified	Plan/region: confirm
SQLite	Default local database	Local file; not cloud hosted by default
Clerk	Authentication for normal mode	Hosted service; region/plan: confirm

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Claude Opus 5.5 (configured default; optional)	Anthropic	Pre-trained AI model; app parser calls it only when LLM_ENABLED=true. Disabled by default; confirm if actually called.	Anthropic API/model terms; confirm exact model/version used.
OpenAI Codex	OpenAI	AI-assisted development tool used by the team; not embedded in Glacier Link.	OpenAI service terms; confirm account/model.
Replit	Replit	Development/cloud platform used to connect and run the frontend UI with the FastAPI backend; not an AI model.	Replit service terms; confirm plan and deployment status.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Glacier Link demo world	Project seed/demo materials	Synthetic	Team-created; verify	N
Fault-message parser evaluation samples	artifacts/api-server/python/tests/data/fault_messages.csv	Synthetic/test examples; verify provenance	Team-created or source to confirm	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Operational records in live use	User/staff-entered via app	Private/user-provided; prototype schema	No external dataset license; retention policy to confirm	Potentially Y

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No — Codex/ChatGPT assistance is evident in the drafting and prior code-editing workflow; confirm all tools used by every member.

OpenAI Codex/ChatGPT; add any other code assistants, image generators or writing tools used by the team.

Tool & Version	Used For	Extent of Use (minor / moderate / major)
OpenAI Codex / ChatGPT (confirm team usage)	Code assistance, UI/text edits and proposal drafting	Moderate; team must confirm scope
Other assistants / image tools	None identified from inspected repository; team to disclose any use	To confirm

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

The team reviewed AI-assisted changes against the codebase, requirements and demo workflow, and remains responsible for the submitted code and claims. AI output must be checked by a team member, tested where relevant, and edited to match actual implementation. Confirm testing/review practices and disclose each tool used by every member before signing.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No, based on the materials reviewed — team must verify against event-provided resources.

None confirmed. If any sponsor API, credits, platform, dataset or mentoring was used, name it and explain the use here.

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
SDC Hackathon 2026 / XMUM AI Club	Replit sponsorship / platform access	Used to connect and run the React frontend UI with the FastAPI backend for the Glacier Link prototype. Confirm whether this includes credits and provide the deployment URL/plan if applicable.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

The Replit sponsorship supported connecting and running the frontend and backend during development. Replit could be replaced by another hosting platform, but the prototype would need to be reconfigured and redeployed there. The core Glacier Link incident-management logic is in the application code and does not depend on Replit-specific functionality.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

The demo uses synthetic staff, business, equipment and incident records. A real deployment may store names, account identifiers, staff fault reports, equipment readings, incident history and operational messages; report text could contain personal details. Clerk handles authentication in normal mode. The optional Anthropic parser receives report text only when enabled; do not send secrets or unnecessary personal data. Demo mode simulates users. The inspected materials do not specify retention, deletion, encryption-at-rest or production access controls; confirm these before deployment and describe the actual hosting protections and retention period.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

Claude may misinterpret ambiguous reports, generate unsupported details, become unavailable, apply rate limits or incur usage costs. It is optional and disabled by default; deterministic rules provide a fallback, and temperature severity, approvals, permissions and recovery remain rule-based. Replit can introduce deployment or availability limits, configuration differences and dependence on its hosting environment; another host would require reconfiguration, and persistent storage must be verified. Codex may suggest incorrect, insecure or unsuitable code or text, so team members must review and validate its output rather than rely on it automatically. Rules can also miss unusual wording or incorrect sensor inputs, so users should verify the unit and reading. The prototype uses synthetic data and webpage notifications; production reliability and real-world waste reduction have not yet been established.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



Date: 24/10/2026